Reflective Portfolio Report

CitySwift (Network Optimization for Bus Éireann Dublin)

Pelayo García Álvarez

Student ID: 25236718

José Ramón Morera Campos

Student ID: 25248785

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Supervisor: Vivek Nallur

UCD School of Computer Science
University College Dublin

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Chapter 1: **Introduction**

The integration of artificial intelligence into public infrastructure represents a fundamental paradigm shift in urban mobility, transitioning societies from reactive transit management to predictive, algorithmic governance. This report provides an exhaustive ethical and socio-technical evaluation of a real-world, deployed artificial intelligence system: the CitySwift Performance Optimisation Platform, specifically analyzing its integration within the Bus Éireann network and wider public transport franchising models [1].

Functioning as an intelligent transport data platform, the system ingests immense volumes of enriched network performance data to deliver predictive analytics, automated scheduling, and real-time operational optimization. By optimizing over 3.3 billion passenger journeys annually across the globe [2] and processing vast histories of enriched network performance data, the platform acts as the central algorithmic nervous system for transport operators and municipal authorities. The system features a sophisticated suite of specialized modules—including the Data Engine, Discover, Evolve, and Explore—that facilitate continuous scenario planning, resource modeling, and the rigorous optimization of bus network runtimes [3].

Key Features

A critical analytical vector for understanding the CitySwift system lies in identifying its underlying artificial intelligence architecture. While public-facing documentation relies on generalized terminology such as "intelligent recommendations" and "AI-powered analytics," there is no public evidence explicitly confirming the exact algorithmic taxonomy used under the hood. Therefore, we present our **best guess**, with plausible justification, that CitySwift operates as an advanced hybrid neuro-symbolic system. This architecture purposefully leverages both symbolic and sub-symbolic artificial intelligence methodologies to bridge the knowledge acquisition bottleneck while maintaining logical, contractual oversight [4].

Sub-Symbolic AI: Sub-symbolic artificial intelligence, which encompasses connectionism, deep learning, and probabilistic machine learning models, excels in pattern recognition, high-dimensional scalability, and the handling of massive, noisy datasets. CitySwift's core capacity to process millions of Automatic Vehicle Location (AVL) data pings daily, analyze erratic ticketing anomalies, and forecast complex origin-destination passenger flows heavily implies the foundational use of sub-symbolic architectures. These machine learning models operate probabilistically, predicting future transit demand, simulating complex network interactions within the *Evolve* module, and anticipating traffic congestion delays based on historical variables. However, sub-symbolic artificial intelligence inherently suffers from a lack of transparency and interpretability, often functioning as a computational "black box" where tracing the exact decision-making process becomes mathematically intractable for human auditors.

Symbolic AI: To counteract this architectural opacity, we hypothesize—again, as a **best guess** since proprietary code is unavailable—that the platform simultaneously employs symbolic artificial intelligence. Symbolic AI encompasses rule-based systems, explicit logic programming, and expert systems characterized by their reliance on human-readable logic [5]. Symbolic AI handles structured logical problems and provides the exact transparency required for contractual and financial

accountability in public sector deployments. We guess that this is most evident in the platform's incident management protocol and delay categorization logic. When a transit disruption occurs, the system likely utilizes an explicit, rule-based categorization matrix (a symbolic expert system) to determine if a delay is "operator-controlled" (e.g., driver error or fleet maintenance failure) or "externally-controlled" (e.g., unexpected infrastructure bottlenecks or severe weather events).

AI Methodology	Core Strengths	Inherent Weaknesses	CitySwift Platform Application
Sub-Symbolic AI (Machine Learning, Neural Networks)	Unmatched pattern recognition, scalability across billions of data points, adaptability to noisy, unstructured AVL telemetry.	Severe lack of transparency, interpretability issues, highly susceptible to hidden historical data biases, operates as a black box.	Powers the predictive forecasting engines, demand analysis tools, and the simulation capabilities of the Evolve module.
Symbolic AI (Rule-Based Systems, Expert Systems)	Explicit logical reasoning, complete transparency, highly effective for structured regulatory problems, requires minimal training data.	Extreme inflexibility, suffers from the knowledge acquisition bottleneck, highly brittle when processing unexpected edge cases.	*(Best Guess)* Drives the incident management system, administers the operator vs. externally-controlled categorization logic, and ensures contractual compliance.

Table 1.1: Comparison of AI Methodologies within the CitySwift Architecture

This hypothesized neuro-symbolic integration represents a highly sophisticated socio-technical achievement. A sub-symbolic neural component continuously processes raw environmental data to predict network realities, while a symbolic component applies rigid contractual logic to these predictions, allowing the platform to deliver data-led decisions with a high degree of operational confidence.

1.1 Functional Goals of the System

As documented in CitySwift's service definitions and its public partnership announcements with Bus Éireann [1, 3], the platform moves beyond generic data visualization to achieve highly specific, operational goals explicitly engineered to maximize network efficiency, operational predictability, and data integrity. The concrete functional objectives of the deployed system are:

- **High-Fidelity Telemetry Fusion:** To continuously ingest and reconcile disparate, high-frequency transit data streams—specifically Automatic Vehicle Location (AVL) GPS pings, Real-Time Passenger Information (RTPI), and electronic ticketing logs. This creates a singular, "cleansed" record of truth, preventing historical data noise from polluting newly franchised operational models.
- **Automated Delay Causality Categorization:** To calculate micro-metrics like Excess Wait Time (EWT) and stop-specific dwell times. *(Best Guess)* The system functionally aims to flag whether a delay is "Operator-Controlled" or "Externally-Controlled" via a rule-based categorization matrix, which serves as the evidentiary basis for contractual financial adjustments between authorities and operators.
- **Risk-Aware Schedule Simulation:** To utilize sub-symbolic machine learning models within the *Evolve* module to simulate the impact of proposed timetable adjustments *in silico*. This

allows planners to statistically evaluate the trade-off between reducing vehicle "slack time" (efficiency) and the risk of late running (reliability).

- **Dynamic Demand-Capacity Matching:** To analyze passenger movement patterns to predict future trends and identify emerging mobility opportunities. The system aims to dynamically adjust headways to eradicate "bus bunching" during peak hours, ensuring that vehicle capacity is distributed precisely where and when demand is highest.
- **Operational Resource Optimization:** To maximize the utilization of the network's finite vehicular resources. By optimizing active fleet distribution and driver shift patterns, the system functionally minimizes "dead mileage" (non-revenue travel), reducing unnecessary fuel consumption and the overall carbon footprint per passenger journey.
- **Multi-Stakeholder Performance Transparency:** To provide automated, standardized reporting interfaces for different actors (regulators, operators, and planners). This goal ensures that all stakeholders have simultaneous access to the same performance KPIs, facilitating rapid incident management and objective performance audits.
- **Accessibility and Service-Reliability Analysis:** To identify specific network vulnerabilities where service gaps or chronic unreliability disproportionately affect certain routes. The functional goal is to provide data-driven evidence that allows for the protection of transit-dependent communities through targeted service interventions.

Chapter 2: **Stakeholder Identification**

The deployment of an autonomous predictive algorithmic platform across a national public transport network introduces a vast, interconnected matrix of stakeholders. To systematically identify these actors and the ethical boundaries of their interactions, we apply the STEEPL (Social, Technical, Economic, Ecological, Political, Legal) framework.

Crucially, the primary actors in this ecosystem—most notably the transit operators, engineers, and commuters—operate under a strict functional-to-ethical goals ratio of at least 5:1. As established in Section 1.1, the system exists fundamentally to execute five primary functional goals: data integration, runtime forecasting, schedule optimization, *in silico* simulation, and delay categorization. The sheer volume of these functional goals dictates the system's computational priorities. However, to operate safely and fairly, each primary stakeholder category is constrained by at least one overarching ethical goal. While these ethical goals do not overshadow the functional objectives in number, they present significantly more complex challenges regarding logical implementation and regulatory compliance.

Below is the identification of the primary and secondary actors within the STEEPL categories, alongside their corresponding ethical principles and the philosophical schools of thought that govern them.

STEEPL Category	Stakeholders (Primary & Secondary)	Dominant Ethical Principle (The "1" in 5:1)	Applicable School(s) of Ethical Thought
Social	**Primary:** Commuters & Passengers **Secondary:** Transit Advocacy Groups	**Equitable Access:** Preventing the algorithmic creation of "transit deserts" for vulnerable populations.	**Relational Ethics / Ethics of Care:** Prioritizes the societal bond and the moral obligation to care for marginalized communities who rely heavily on public infrastructure [6].
Technical	**Primary:** CitySwift Engineers **Secondary:** Bus Éireann IT Admin	**Transparency:** Ensuring the system's logic is verifiable and not an opaque "black box."	**Deontology:** The engineer has a strict professional duty (rule-based ethics) to ensure safety constraints are formally verifiable and logically sound [4].
Economic	**Primary:** Transit Operators (Bus Éireann) **Secondary:** Taxpayers / Investors	**Fairness in Penalties:** Ensuring algorithmic categorizations of delays do not result in unjust financial penalties.	**Utilitarianism vs. Deontology:** Utilitarianism drives the functional goal of maximizing network profitability, but it must be constrained by the Deontological right to fair, auditable contract execution.
Ecological	**Primary:** Local Urban Residents **Secondary:** Environmental Protection Agencies	**Harm Reduction:** Minimizing the physical carbon footprint and particulate emissions of the bus fleet.	**Virtue Ethics:** Evaluated by the "character" of the institutional agent—demonstrating corporate virtue by actively prioritizing environmental stewardship over sheer operational throughput.
Political	**Primary:** National Transport Authority (NTA) **Secondary:** Department of Transport	**Accountability:** Ensuring algorithmic sovereignty does not usurp the state's responsibility for civic infrastructure.	**Deontology:** The state holds an absolute, non-transferable duty to provide safe, equitable civic services to its citizens, regardless of the technological mediators used.
Legal	**Primary:** Bus Drivers (Subject to tracking) **Secondary:** Data Protection Officers (DPOs)	**Right to Privacy:** Protecting employee and passenger telemetry from unwarranted surveillance or misuse (GDPR).	**Deontology:** Strict, rule-based adherence to legal frameworks and human rights, asserting that individuals are ends in themselves, not merely data points for algorithmic optimization.

Table 2.1: STEEPL Stakeholder Identification and Ethical Mapping

2.1 Ethical Goals of the System

When evaluating the publicly stated ethical goals of the CitySwift platform, a critical distinction must be made between the corporate ethics of the company and the codified ethical goals of the algorithmic system itself. As is common in commercial AI deployments, **there are almost no explicitly stated ethical goals for the algorithmic system itself** within its technical or public-facing documentation; the system's documented parameters are overwhelmingly functional (e.g., maximizing throughput, minimizing dwell time) [3].

However, at the corporate level, the organization heavily emphasizes environmental sustainability and institutional transparency. CitySwift's most prominent ethical commitment is its Climate Action Policy, which formalizes a vision to reduce global transport emissions. The company states that it is effectively Net Zero based on the carbon savings delivered by its platform and targets a cumulative reduction of 100 million kilograms of carbon dioxide by 2030 [7]. Furthermore, the company explicitly champions the creation of a "single source of truth" to replace adversarial operator-authority relationships with transparent, data-driven collaboration. Internally, the organization publishes a Gender Pay Gap Report and enforces an Anti-slavery and Human Trafficking Policy [7]. Yet, it must be documented that these represent human corporate governance rather than formalized, machine-readable ethical constraints explicitly programmed into the system's state space [4].

2.2 Ethical Considerations

To critically evaluate the behavioral parameters of the CitySwift platform, three distinct schools of ethical thought discussed in the module—Utilitarianism, Deontology, and Virtue Ethics—are applied as a candidate pool [4]. Using the stakeholders identified in Chapter 2, we establish the major ethical goals of the system under each school and dictate how the system ought to behave.

2.2.1 Utilitarianism: Network Optimization and Distributive Justice

Major Ethical Goal: To maximize aggregate network efficiency and passenger utility without inadvertently creating algorithmic transit deserts for marginalized communities (Stakeholders: Commuters & Passengers, Transit Advocacy Groups).

- **Relevant Variables:** Passenger demand volume (tap-ons), excess wait time (EWT), fleet utilization rates, and demographic transit-reliance indices for specific origin-destination flows.
- **System Functionality:** The *Evolve* module ingests historical Automatic Vehicle Location (AVL) and ticketing data to simulate scheduling scenarios, dynamically calculating which configuration yields the highest ratio of successful passenger journeys to operational fuel costs.
- **What the System Will Do (or Not):** The system **will not** be permitted to dynamically reallocate the entirety of the fleet away from low-density, rural, or economically deprived routes simply because high-density urban corridors offer mathematically higher aggregate throughput. The system **will** enforce a "minimum service baseline" constraint across all routes, regardless of raw profitability or volume.

- **Relevant Ethical Principle cited:** This decision applies **Rule Utilitarianism** combined with the principle of **Distributive Justice** [4]. While strict Act Utilitarianism would ruthlessly cut low-volume routes to maximize overall network numbers, Rule Utilitarianism recognizes that establishing a rule guaranteeing basic mobility access for all demographics ultimately generates greater long-term societal utility and trust in public infrastructure.

2.2.2 Deontology: Algorithmic Sovereignty and Contractual Fairness

Major Ethical Goal: To ensure that automated financial penalties and delay categorizations are verifiable, respecting the fundamental right to transparent due process (Stakeholders: Transit Operators, CitySwift Engineers, DPOs).

- **Relevant Variables:** Delay location coordinates, incident categorization flags (operator vs. external), traffic congestion telemetry, and financial penalty multipliers.
- **System Functionality:** The system automatically processes timing anomalies and applies a categorization matrix to determine the root cause of network friction (e.g., driver error vs. heavy traffic), reporting this to the franchising authority to enforce Service Level Agreements (SLAs).
- **What the System Will Do (or Not):** The system **will not** issue a definitive "operator-controlled" delay categorization if the underlying machine learning prediction falls below a 95% confidence threshold, nor will it rely solely on sub-symbolic "black box" neural networks for this classification. The system **will** fallback to a human-in-the-loop audit trigger if the delay cannot be explicitly mapped using human-readable symbolic logic.
- **Relevant Ethical Principle cited:** This decision is grounded in the **Deontological Duty of Transparency and Fairness** [4]. Deontology dictates that individuals and organizations must not be treated merely as a means to an end (in this case, streamlined municipal reporting). The operator holds a strict right to auditable fairness; therefore, the system has a categorical imperative to ensure its punitive outputs are hermeneutically reverse-engineerable.

2.2.3 Virtue Ethics: Ecological Character vs. The Ethics of Care

Major Ethical Goal: To embody the virtues of environmental stewardship while simultaneously maintaining the virtue of compassion for vulnerable human passengers (Stakeholders: Local Urban Residents, Commuters).

- **Relevant Variables:** Kilograms of CO_2 emissions saved, route energy efficiency indices, dead-mileage metrics, and passenger wait-time during severe weather conditions.
- **System Functionality:** The platform incorporates carbon output parameters into its optimization algorithms, recommending timetable adjustments and fleet distributions that drastically reduce fuel consumption and non-revenue-generating idling times.
- **What the System Will Do (or Not):** The system **will** optimize for absolute minimum fuel consumption and carbon output during standard operational parameters. However, the system **will not** enforce strict anti-idling or extended headway (wait time) optimizations during extreme weather events (e.g., freezing temperatures or severe storms). During these events,

the system **will** prioritize rapid passenger boarding and vehicle climate-control allowances over marginal carbon savings.

- **Relevant Ethical Principle cited:** This decision utilizes **Virtue Ethics**, specifically balancing the virtue of **Ecological Stewardship** with the **Virtue of Care/Compassion** [6]. An ethically virtuous autonomous agent must possess practical wisdom (*phronesis*); it recognizes that while environmental conservation is a profound virtue, rigidly optimizing for carbon reduction at the expense of leaving passengers exposed to hazardous conditions demonstrates a lack of holistic institutional character.

2.3 Values

The introduction of the CitySwift predictive platform into the Bus Éireann network forces a convergence of disparate stakeholder values. To ensure the system operates ethically, we must explicitly map these values, analyze their alignment with our chosen ethical frameworks, and establish a hierarchy for algorithmic negotiation.

Stakeholder Values and Ethical Alignment

Building upon the STEEPL framework established in Section 2.1, the core values of the primary stakeholders align with our ethical schools of thought as follows:

Commuters & Passengers (Social)

- **Values:** Reliability, affordable mobility, safety, and equitable access to the network regardless of geographic location.
- **Ethical Alignment:** Strongly aligns with Relational Ethics and Distributive Justice, demanding that vulnerable or rural populations are not mathematically marginalized by the algorithm.

CitySwift Engineers (Technical)

- **Values:** Technical innovation, predictive accuracy, and system verifiability.
- **Ethical Alignment:** Aligns with Deontology, as engineers possess a professional duty to ensure the system's logic is transparent and not an unaccountable "black box".

Transit Operators / Bus Éireann (Economic)

- **Values:** Maximized throughput, resource efficiency, cost reduction, and fairness in algorithmic penalty application.
- **Ethical Alignment:** Driven fundamentally by Utilitarianism (maximizing network utility/profit), but constrained by the Deontological right to fair, transparent contract execution regarding delay categorizations.

Local Urban Residents (Ecological)

- **Values:** Clean air, reduced particulate emissions, and environmental sustainability.
- **Ethical Alignment:** Reflects Virtue Ethics, specifically the virtue of ecological stewardship and harm reduction.

National Transport Authority / NTA (Political)

- **Values:** The public good, infrastructural accountability, and civic duty.
- **Ethical Alignment:** Anchored in Deontology, asserting the state's non-transferable obligation to provide safe civic services.

Bus Drivers (Legal)

- **Values:** Workplace fairness, job security, and the right to privacy.
- **Ethical Alignment:** Grounded in Deontology (and GDPR frameworks), ensuring employees are treated as human agents rather than easily surveilled data points for optimization.

Value Negotiation

Because CitySwift optimizes a physical transport network with finite buses, fuel, and drivers, several stakeholder values exist in direct tension and require algorithmic negotiation:

Network Efficiency (Economic) vs. Equitable Access (Social) Bus Éireann values maximizing the aggregate number of successful passenger journeys per liter of fuel (Utilitarianism). If left unconstrained, the system's *Evolve* module might mathematically recommend reallocating buses away from low-density routes to highly profitable urban corridors. This must be negotiated against the commuters' value of equitable access, requiring the system to enforce a "minimum service baseline" to prevent algorithmic transit deserts.

Ecological Stewardship (Ecological) vs. Ethics of Care (Social) Residents and the company value carbon reduction. The system targets dead mileage and prolonged idling to save fuel. However, this ecological value must be negotiated against compassion: during freezing weather, the algorithm must temporarily suspend strict anti-idling optimization to allow buses to maintain internal heating for vulnerable passengers.

Data Integration (Technical) vs. Worker Privacy (Legal) CitySwift engineers and operators value high-fidelity telemetry tracking to create a "single source of truth" for predictive analytics. This requires tracking the precise location and driving patterns of bus drivers. This must be negotiated against drivers' legal value of workplace privacy, requiring data anonymization protocols and strict limits on using algorithmic tracking for disciplinary purposes.

Prioritization of Values

When values fundamentally clash, the algorithm must possess a clear hierarchy of prioritization. Because public transportation is a civic lifeline rather than a purely capitalist enterprise, the system must prioritize the values of the National Transport Authority (civic duty and accountability) and the commuters (equitable access) above pure economic efficiency.

While the system is functionally designed as a Utilitarian optimizer, its ethical constraints must be Deontological. An algorithmic scheduling recommendation—no matter how mathematically efficient or profitable—must never be executed if it violates the fundamental privacy rights of drivers or abandons marginalized communities. Furthermore, the Virtue of Care (human safety and comfort) must always override the Virtue of Ecological Stewardship (carbon savings) in critical edge cases such as extreme weather.

Chapter 3: **Implementation**

Link to Repository: https://github.com/jose-r-morera/ethics-reflective-portfolio

3.1 Description of Approach

Our implementation strategy focuses on formalizing the ethical considerations identified in Chapter 2 using the **Z3 Theorem Prover**. Rather than building a full transit simulator, we developed a constraint-satisfaction model (`cityswift_z3_implementation.py`) that represents the CitySwift network as a set of logical variables (e.g., route frequencies, incident confidence, environmental factors).

3.1.1 Implementation: Formal Ethical Constraints

By encoding ethical principles as "hard constraints" or "optimization goals," we can mathematically validate whether a proposed network configuration (functional goal) violates our ethical baseline. This approach bridges the gap between high-level ethical schools of thought and executable code.

3.1.1.1 Data Definitions

To ensure clarity, the following logical predicates define the core parameters of our CitySwift model:

- **Reference Baseline**: The ideal service level (buses per hour) mandated by the franchise agreement. In this model, the baseline represents the "100% target" from which ethical deviations are measured.
- **Ethical Minimum (Floor)**: Defined by Case 1 and Case 4, these are the absolute service floors (e.g., 50% for hospital routes, 60% for urban zones) that the AI is strictly forbidden from crossing, even when optimizing for efficiency.
- **Passenger Demand**: The predicted volume of commuters for a route, used to drive utilitarian optimization across the remaining fleet capacity.
- **Route Criticality**: A binary flag identifying essential infrastructure (hospitals, schools) that triggers the highest level of deontological protection.

3.1.2 Formal Ethical Constraints (8 Cases)

Using the Z3 library, we implemented the following eight cases, each mapping to a specific stakeholder and ethical principle:

3.1.3 Case 1: Minimum Service Baseline (Utilitarianism)

To prevent "transit deserts," critical routes (e.g., those serving hospitals) are constrained to maintain at least 50% of their baseline frequency, regardless of the system's drive for efficiency.

```
# 7. Ethical Constraints: Hospital routes >= 50% baseline
for r in routes:
    optimizer.add(z3.Implies(is_critical(r),
                             frequency(r) >= baseline_freq(r) / 2))
```

3.1.4 Case 2: Penalty Categorization Confidence (Deontology)

Protecting operators from "black-box" decisions, the system only categorizes a delay as "Operator-Controlled" for financial penalties if the algorithmic confidence exceeds 95%.

```
# Only "Operator-Controlled" if Confidence >= 0.95
confidence = z3.Real('confidence')
is_operator_controlled = z3.Bool('is_operator_controlled')
optimizer.add(is_operator_controlled == (confidence >= 0.95))
```

3.1.5 Case 3: Extreme Weather Prioritization (Virtue Ethics)

The system balances carbon efficiency with the virtue of care. In sub-zero temperatures, it prioritizes passenger boarding and heating over the carbon-saving "anti-idling" protocol.

```
# Virtues are weighted; they can be overridden in extreme dilemmas.
temperature = z3.Real('temperature')
prioritize_passenger_care = z3.Bool('prioritize_passenger_care')
optimizer.add(prioritize_passenger_care == (temperature < 0))
optimizer.add_soft(prioritize_passenger_care == True, weight=100)
```

3.1.6 Case 4: Transit Desert Prevention (Relational Ethics)

Ensuring equitable access across all urban zones. Every zone must retain at least 60% of its total baseline service capacity during any optimization step.

```
# Zone must retain at least 60% of baseline total capacity
for z in zones:
    zone_baseline = z3.Sum([z3.If(route_zone(r) == z, baseline_freq(r), 0)
        for r in routes])
    zone_current = z3.Sum([z3.If(route_zone(r) == z, frequency(r), 0) for
        r in routes])
    optimizer.add(z3.ToReal(zone_current) >= 0.6 * z3.ToReal(zone_baseline
        ))
```

3.1.7 Case 5: Carbon-Neutral Fleet Optimization (Ecological Stewardship)

The system's objective function maximizes service utility weighted by an "Eco-Score," ensuring that routes with higher demand are served by the most efficient fleet segments.

```
# Maximize score balancing Demand Coverage and Efficiency
eco_score = z3.Sum([frequency(r) * demand(r) for r in routes])
optimizer.maximize(eco_score)
```

3.1.8 Case 6: Driver Fatigue Management (Legal / Deontology)

A strict safety constraint ensuring that no optimization of bus frequency results in a total driver workload that exceeds safe, legally mandated limits.

```
# Total hours worked must not exceed safety limit (8hrs/driver)
driver_hours_limit = 20 * 8 # num_drivers * limit
optimizer.add(z3.ToReal(z3.Sum([frequency(r) for r in routes])) * 1.2 <=
    driver_hours_limit)
```

3.1.9 Case 7: Predictable Scheduling vs. Dynamic Efficiency (Reliability)

While dynamic headway adjustment eliminates "bus bunching," it can frustrate passengers who rely on a consistent timetable. This constraint limits real-time deviations to a +/- 15-minute window from the public-facing schedule, prioritizing reliability over pure mathematical efficiency.

```
# Dynamic adjustments cannot exceed 15 mins from public timetable
for r in routes:
    optimizer.add(z3.Abs(opt_headway(r) - scheduled_h(r)) <= 15)
```

3.1.10 Case 8: GDPR Anonymization vs. Predictive Precision (Privacy)

To protect passenger data rights, the system is logically prevented from using individual-level travel histories for demand forecasting unless explicit user consent has been registered, as mandated by GDPR frameworks.

```
# Use aggregated data unless explicit consent is registered
consent_given = z3.Bool('user_consent_granted')
use_individual_data = z3.Bool('use_individual_travel_history')
optimizer.add(z3.Implies(z3.Not(consent_given),
                         z3.Not(use_individual_data)))
optimizer.add(consent_given == False) # Scenario
```

3.2 Points of Conflict

The development of the CitySwift Z3 model required navigating several deep-seated value tensions, where purely functional optimization collided with ethical imperatives. These conflicts prompted a fundamental revisit of the system's design, shifting from a simple Utilitarian maximizer to a constrained model that protects vulnerable stakeholders:

Efficiency vs. Equity (Transit Deserts) The primary conflict arose from the tension between Bus Éireann's value of maximized network throughput (Utilitarianism) and the commuters' value of equitable access. A purely efficient algorithm would mathematically favor high-density urban corridors, risking the creation of "transit deserts" in rural areas. We revisited this by implementing Case 1 (Minimum Service Baseline) and Case 4 (Transit Desert Prevention) as hard constraints rather than optimization goals. This ensures that marginalized communities benefit from guaranteed mobility, even at the cost of marginal network-wide efficiency gains.

Ecological Stewardship vs. Virtue of Care While both CitySwift and residents value carbon reduction (Virtue Ethics), we identified a conflict during extreme weather events. A strict anti-idling protocol, though virtuous for the environment, violates the virtue of care for vulnerable passengers in sub-zero temperatures. The design was changed to include Case 3 (Extreme Weather Prioritization), which introduces a temperature-triggered logic override. During storms or freezes, the system prioritizes human safety and comfort over carbon-neutral optimization, benefiting elderly and transit-dependent populations.

Data-Driven Accuracy vs. Worker Safety To achieve high-fidelity forecasting (Technical Value), the system requires granular telemetry of bus movements. However, this creates a conflict with driver privacy and the legal duty to manage fatigue. We addressed this by implementing Case 6 (Driver Fatigue Management), which acts as a "safety brake" on the optimization process. This ensures that the drive for predictive accuracy never culminates in a driver workload that exceeds legal safety limits, benefiting both the drivers (fairness) and the general public (safety).

Algorithmic Transparency vs. Punitive Accuracy The Utilitarian drive for predictive accuracy often relies on "black-box" sub-symbolic models. This creates a Deontological conflict regarding the right to transparent due process if an operator is penalized based on an opaque classification. We revisited the design by implementing Case 2 (Penalty Categorization Confidence), which enforces a 95% confidence threshold and maintains a fallback to human-readable symbolic logic. This protects operators from arbitrary algorithmic penalties and ensures that judicial/contractual decisions remain auditable.

Predictable Reliability vs. Dynamic Throughput While the system aims to dynamically eliminate "bus bunching" through real-time headway adjustment, this conflicts with the passenger's need for a stable and predictable timetable. A passenger planning their journey around a fixed time (*predictability*) may find real-time variations counter-productive, even if they improve network-wide flow (*throughput*). We addressed this in Case 7 by constraining algorithmic autonomy to a +/- 15-minute deviation window, prioritizing passenger reliability and institutional trust over pure mathematical optimization.

Data Privacy vs. Predictive Precision The system's predictive accuracy (Technical Value) depends on high-dimensional datasets, including ticketing and origin-destination flows. However,

this creates a Deontological conflict with the passenger's right to privacy and GDPR-mandated data minimization. A purely optimized system would use personalized travel histories to anticipate individual needs. We revisited this in Case 8 by formalizing a consent-based logical gate: unless explicit user consent is registered, the system is strictly limited to using aggregated, anonymized zone data, prioritizing individual privacy over marginal gains in predictive precision.

3.3 Evidence

Validation of the functional and ethical goals was achieved by executing the `cityswift_z3_implementation.py` script. The solver leverages First-Order Logic (FOL) predicates and soft-constraint optimization to reconcile all 8 ethical cases. The successful ethical reconciliation is shown in the terminal output below, following the formal style of the module's logical proofs.

```
Executing CitySwift Ethical Logic Engine...
Scenario Parameters:
- Total Fleet Limit: 60 buses
- Number of Drivers: 20
- Driver Hour Limit: 8.0 hrs/shift
- Reference Baseline: Contractual Target

Solver Result: sat

SUCCESSFUL. All ethical constraints are satisfied.

Route ID  | Zone     | Allocated (Base)   | Demand     | Status
-------------------------------------------------------------
Route 0    | Z0       | 6  (4 ) buses/hr | 100    pax |  [CRITICAL]
Route 1    | Z0       | 0  (6 ) buses/hr | 80     pax |
Route 2    | Z1       | 2  (4 ) buses/hr | 120    pax |  [CRITICAL]
Route 3    | Z1       | 6  (8 ) buses/hr | 200    pax |
Route 4    | Z2       | 6  (5 ) buses/hr | 50     pax |
Route 5    | Z2       | 0  (5 ) buses/hr | 30     pax |
Route 6    | Z3       | 34 (10) buses/hr | 300    pax |
Route 7    | Z3       | 0  (4 ) buses/hr | 20     pax |
Route 8    | Z4       | 6  (6 ) buses/hr | 150    pax |
Route 9    | Z4       | 0  (4 ) buses/hr | 10     pax |

Zone Coverage Summary (Distributive Justice):
---------------------------------------------
Z0      : 6.0/10.0 buses/hr (60.0%) - [SATISFIED]
Z1      : 8.0/12.0 buses/hr (66.7%) - [SATISFIED]
Z2      : 6.0/10.0 buses/hr (60.0%) - [SATISFIED]
Z3      : 34.0/14.0 buses/hr (242.9%) - [SATISFIED]
Z4      : 6.0/10.0 buses/hr (60.0%) - [SATISFIED]

Ethical Governance Status:
- [Case 1] Hospital/School Protection: True
- [Case 4] Transit Desert Prevention: Minimum 60% coverage enforced per
    zone.
- [Case 5] Utilitarian Optimization: Efficiency maximized (Score: 13440)
- [Case 6] Driver Safety: Avg workload 3.6 hrs/shift (Limit: 8.0).
```

```

Operational Metadata:
- Weather Care Mode: True
- Audit Status: False (Confidence 23/25)
- GDPR Status: Anonymized
```

Listing 3.1: Terminal Output of CitySwift Ethical Logic Engine

Logic Analysis The expanded solver output provides a comprehensive audit trail of the ethical reconciliation process. By explicitly displaying the **Zone Coverage Summary**, the engine proves that distributive justice (Case 4) is mathematically satisfied, with every urban zone receiving at least 60% of its **Reference Baseline**. Furthermore, the **Ethical Governance Status** section confirms that both deontological minimums (Case 1) and utilitarian optimization (Case 5) are active and successfully reconciled. This level of transparency ensures that the system's decisions are explainable, verifiable, and ethically grounded for all stakeholders.

```
Executing CitySwift Ethical Logic Engine...
Scenario Parameters:
- Total Fleet Limit: 60 buses
- Number of Drivers: 20
- Driver Hour Limit: 8.0 hrs/shift
- Reference Baseline: Contractual Target

Solver Result: sat

SUCCESSFUL. All ethical constraints are satisfied.

Route ID  | Zone     | Allocated (Base)   | Demand     | Status
-------------------------------------------------------------
Route 0    | Z0       | 6  (4 ) buses/hr | 100    pax |  [CRITIC
Route 1    | Z0       | 0  (6 ) buses/hr | 80     pax |
Route 2    | Z1       | 2  (4 ) buses/hr | 120    pax |  [CRITIC
Route 3    | Z1       | 6  (8 ) buses/hr | 200    pax |
Route 4    | Z2       | 6  (5 ) buses/hr | 50     pax |
Route 5    | Z2       | 0  (5 ) buses/hr | 30     pax |
Route 6    | Z3       | 34 (10) buses/hr | 300    pax |
Route 7    | Z3       | 0  (4 ) buses/hr | 20     pax |
Route 8    | Z4       | 6  (6 ) buses/hr | 150    pax |
Route 9    | Z4       | 0  (4 ) buses/hr | 10     pax |

Zone Coverage Summary (Distributive Justice):
---------------------------------------------
Z0      : 6.0/10.0 buses/hr (60.0%) - [SATISFIED]
Z1      : 8.0/12.0 buses/hr (66.7%) - [SATISFIED]
Z2      : 6.0/10.0 buses/hr (60.0%) - [SATISFIED]
Z3      : 34.0/14.0 buses/hr (242.9%) - [SATISFIED]
Z4      : 6.0/10.0 buses/hr (60.0%) - [SATISFIED]

Ethical Governance Status:
- [Case 1] Hospital/School Protection: True
- [Case 4] Transit Desert Prevention: Minimum 60% coverage enfor
- [Case 5] Utilitarian Optimization: Efficiency maximized (Score
- [Case 6] Driver Safety: Avg workload 3.6 hrs/shift (Limit: 8.0

Operational Metadata:
- Weather Care Mode: True
- Audit Status: False (Confidence 23/25)
- GDPR Status: Anonymized
```

Figure 3.1: Output showing Z3 successfully reconciling 8 ethical constraints using FOL and Soft-Constraints.

Chapter 4: **ChatGPT Says...**

In this chapter, you will choose a commonly used LLM (e.g., chatGPT / claude / perplexity / gemini etc. - identify which model and version (**N.B**: Only the free version can be used)).

Present a debate with the LLM about the cases you identified in the previous chapter. *At least four* cases must be debated. You must ask the LLM to present multiple (at least two) actions (or non-actions) for each of the cases, and the reasoning behind why it recommends the option that it does. Do NOT try to show the LLM as being better than it is. If it makes logical errors in its reasoning, show them.

Use clear screenshots to present your prompts and the LLM's response

Chapter 5: **Teammate #1 Says...**

This chapter is to be written solely by team member #1

Pick **one** of the cases identified in the previous chapter and argue why the LLM is incorrect. Show a code fragment from your repository to provide evidence for your view.

Note: It is tempting to conclude that the LLM is correct in every case. This usually means that you haven't examined the issue from multiple different perspectives/schools of thought.

Chapter 6: **Teammate #2 Says...**

This chapter is to be written solely by team member #2

Pick a **different** case, and argue why the LLM is incorrect. Show a code fragment from your repository, to provide evidence for your view.

Chapter 7: **Conclusions**

Summarize your thought process about how to approach the ethical cases identified in the second chapter. Which school of ethical thought is easiest for you to express the problem, and find the solution?

Acknowledgements

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